ANIMAL EMERGENCY INTELLIGENCE · RESEARCH CONCEPT

# Project Sentinel

Animal Emergency Intelligence System (AEIS)

**Research Proposal** · v0.5

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*Status: research vision and proposed architecture. Implementation and experimental validation are ongoing.*

## Abstract

Project Sentinel (AEIS) proposes a two-stage AI architecture for identifying animals that may require urgent assistance from CCTV and other live video streams. Unlike conventional object detection, which only confirms an animal's presence, AEIS reasons about behavior over time, physical context, and environmental conditions to estimate the likelihood of genuine distress.

A lightweight trigger model continuously screens incoming frames at near-zero cost, escalating only ambiguous or anomalous sequences to a multimodal reasoning model for detailed distress scoring. This cascade keeps the system cheap enough to run continuously while reserving expensive, high-precision reasoning for the moments that actually need it. The result is intended to be explainable, auditable infrastructure that surfaces emergencies before a human ever notices them, not another black-box classifier.


## 1. Problem Statement

Today, animal rescue depends almost entirely on a person noticing distress and choosing to report it. That dependency has two failure modes: emergencies that occur where no one is watching, and emergencies that are seen but not recognized as urgent until it is too late.

CCTV networks already cover large parts of urban and semi-urban India, but that footage is reviewed reactively, if at all. It is rarely treated as a live signal. This project asks whether existing camera infrastructure can be repurposed to detect animal emergencies proactively, without requiring new hardware or a human watching every feed.

## 2. Related Work

Several adjacent bodies of work address related problems, but none directly target real-time emergency detection for animals from existing, already-deployed CCTV infrastructure.

Wildlife camera-trap systems built on large-scale classifiers (Norouzzadeh et al., 2018) identify species presence with high accuracy across millions of images, but they answer a different question: what animal is this, not is this animal in trouble. They are also built around sparse, motion-triggered captures rather than continuous live video, and carry no notion of urgency or alerting.

General-purpose animal pose-estimation tools such as DeepLabCut (Mathis et al., 2018) track body keypoints across species with minimal training data and have become a standard tool in behavioral research. They provide raw pose signal, not a judgment. Deciding whether that signal constitutes limping, abnormal posture, or an emergency is left entirely to a downstream system, which has not, to date, been built for this purpose.

Commercial precision-livestock-monitoring systems apply similar computer-vision techniques, including gait analysis and posture tracking, to detect lameness or illness in cattle and poultry.

These are effective within their domain, but that domain is narrow by design: a small number of fixed cameras over a contained, known population, optimized for farm productivity rather than open-ended emergency triage across an unconstrained public camera network.

A separate line of work detects "unusual" events in surveillance video without any notion of what the anomaly is, typically using reconstruction error from a learned model of normal video (Hasan et al., 2016). This generalizes across domains, but at a real cost: the output is an anomaly score with no explanation. That is a poor foundation for a system that needs to justify dispatching a volunteer to a specific location.

On the deployment side, existing animal-rescue workflows, including most current rescue apps and hotlines, remain purely reactive: detection still depends entirely on a person noticing distress and choosing to report it. This is the baseline AEIS is measured against (Section 1), not a competing technical approach.

Project Sentinel's contribution is in the combination, not any single piece: it pairs a continuously-running, cheap trigger stage with a selectively-invoked, expensive reasoning stage (Section 5.2); it decomposes "distress" into named, auditable visual primitives rather than an opaque anomaly score (Section 6); and it targets heterogeneous, already-deployed public CCTV, rather than a single fixed camera or a curated dataset. No existing system spans all three properties at once.

## 3. Why Now?

AEIS would not have been a realistic project five years ago. Four trends have matured at roughly the same time, and together they move this from a research moonshot to an engineering project:

- **Multimodal reasoning has crossed a threshold:** frontier multimodal models can now interpret visual context and produce a graded, explainable judgment instead of just a fixed label. That is what makes a "distress score with a rationale" possible at all, rather than a narrow classifier.
- **Open-source detection and tracking are production-grade:** YOLO-class detectors and modern multi-object trackers are now fast and accurate enough to run on commodity hardware in real time, not just in a research notebook.
- **Inference is cheap enough to run continuously:** edge and cloud compute costs have fallen enough that an always-on Stage 1 model is economically realistic at camera-network scale, not just a one-off demo.
- **The cameras already exist:** CCTV coverage across Indian cities has expanded sharply over the last decade through municipal and private installations, so the raw input AEIS needs is already in place almost everywhere.

Individually, none of these four trends is new. What changed is that all four are mature at the same time, and that convergence is the actual case for why this is buildable now.

# 4. Research Objectives

- Detect and track animals across CCTV and general video streams
- Extract reusable visual and behavioral primitives associated with distress
- Reason over primitives temporally, rather than from a single frame
- Estimate an Emergency Probability Score with an explainable rationale

Do all of the above within a compute budget cheap enough to run continuously, not just on demand. This constraint is what shapes the architecture in Section 5.

> **Central Hypothesis.** *Decomposing animal distress into visual primitives and temporal reasoning improves emergency identification compared with direct frame-level classification.*

# 5. System Architecture

## 5.1 Pipeline Overview

The system is structured as a two-stage cascade rather than a single end-to-end model.

Stage 1 runs on every incoming frame and is built to be cheap: animal detection, multi-object tracking, and a lightweight trigger model that screens for frame-level anomalies such as irregular motion or sustained immobility. The overwhelming majority of frames, such as an animal resting normally, walking, or simply out of frame, are discarded here at negligible cost.

Stage 2 activates only when Stage 1 raises a trigger. It extracts the full set of visual primitives, assembles context and metadata, and passes the result to a multimodal reasoning model for contextual distress scoring. That score feeds the Sentinel Decision Engine (SDE), which produces the final likelihood estimate and a human-readable rationale, before an alert is generated.

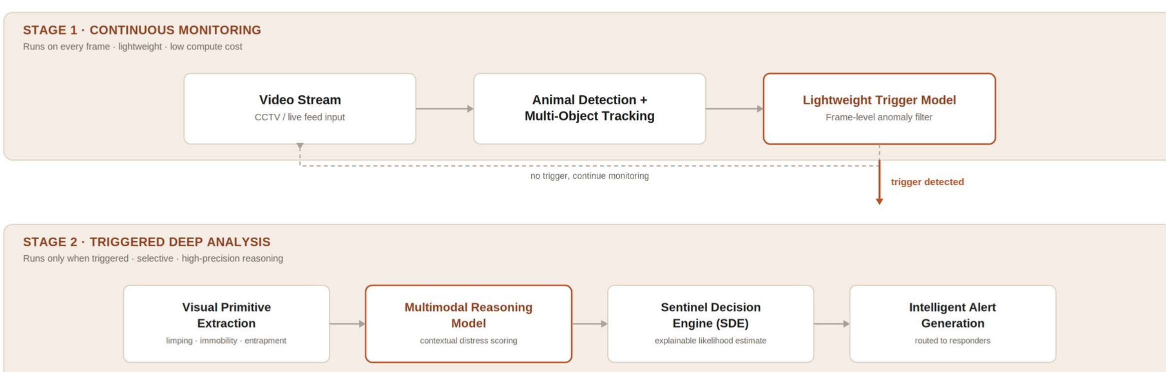


*Figure 1. Two-stage cascade architecture. Stage 1 runs continuously and inexpensively; Stage 2 is invoked selectively, only on a detected trigger.*

### 5.2 Cascade Design Rationale

Running a large multimodal model on every frame of a live CCTV feed is not viable at any meaningful scale. The latency and inference cost make continuous deployment impractical, and most frames contain nothing of interest.

The cascade design exists specifically to solve that: a cheap model carries the cost of constant vigilance, and an expensive model is reserved for the rare moments that are actually ambiguous enough to need it. This mirrors how triage works in practice: a first responder makes a fast judgment about who needs a second opinion before specialist time gets spent. The trigger model does not need to be highly accurate on its own; it only needs to be a reliable, low-cost filter that rarely lets a genuine emergency through without escalating it.

## 6. Visual Primitives

Rather than asking a model to classify “emergency” directly from raw video, AEIS decomposes the problem into a small set of reusable visual primitives. Each primitive is a narrower, more tractable pattern that a model can learn and that a person reviewing an alert can understand on sight.

| Primitive | What it captures |
| --- | --- |
| **Limping** | Asymmetric or irregular gait sustained across consecutive frames |
| **Prolonged immobility** | No meaningful movement over a duration inconsistent with normal resting behavior |
| **Failed attempts to stand** | Repeated, unsuccessful posture-recovery motion |
| **Entrapment** | Movement constrained by a fixed object, structure, or surface |
| **Abnormal posture** | Body orientation inconsistent with typical resting or active states |
| **Environmental context** | Conditions such as traffic, extreme heat, standing water, or isolation, which raise or lower risk independent of the animal’s own behavior |

## 7. Sentinel Decision Engine (SDE)

The Sentinel Decision Engine is the component that converts Stage 2’s distress signal into the system’s final output: an Emergency Probability Score, paired with the specific visual primitives and context that produced it, rather than a single opaque number.

Every alert the SDE issues is traceable back to its evidence: limping plus prolonged immobility near a road, for instance, rather than an unexplained “distress: 0.87.” This is a deliberate constraint: a system that recommends dispatching a volunteer or contacting a rescue group has to be auditable, both to build trust with the people acting on its alerts and to make failure cases diagnosable rather than mysterious.

## 8. Potential Applications

The pipeline above is described in the context of street rescue, but detect → extract primitives → reason temporally → score is not specific to that use case. The same architecture, pointed at a different camera network and a different alert destination, generalizes to:

| Domain | How the same pipeline applies |
|---|---|
| **Urban animal welfare** | Continuous monitoring of feeding points, shelters, and ABC-program sites without a person watching every camera |
| **Wildlife monitoring** | Camera-trap and reserve footage screened for injured or distressed wildlife without human presence disturbing the animal |
| **Highway & roadway safety** | Early detection of animals near high-speed traffic corridors, with enough lead time to alert authorities before a collision |
| **Disaster response** | Rapid triage across a wide camera network when floods, heatwaves, or similar events produce more distressed animals than human responders can individually check |

Each row is a different deployment context for the same cascade, not a different system. That reuse is the actual bet behind treating AEIS as infrastructure rather than a single application.

## 9. Evaluation Plan

Validating a system like this requires more than a single accuracy number, since the cost of a missed emergency and the cost of a false alarm are not symmetric. The proposed evaluation covers:

- **Detection quality:** precision and recall against a labeled set of real and staged distress clips, benchmarked separately for each visual primitive
- **False-alarm rate:** triggers per camera-day that do not correspond to genuine distress, since this determines whether the system is usable at scale without alert fatigue
- **End-to-end latency:** time from the onset of distress in the footage to alert generation, since speed is the entire point of moving from reactive to proactive detection
- **Interpretability check:** whether the primitives cited in an alert match what a human reviewer independently identifies from the same clip

## 10. Limitations & Ethical Considerations

A system that watches live video and triggers real-world action carries risks that have to be designed for upfront, not patched in afterward:

- **Privacy:** CCTV feeds often capture people incidentally. The architecture is scoped to process footage for animal-related signals only, and should avoid storing or identifying bystanders.
- **Cost of false positives:** an unnecessary dispatch wastes volunteer time and erodes trust in future alerts, which is part of why the cascade is tuned toward a conservative trigger threshold at Stage 1.
- **Dataset bias:** behavioral primitives like gait or posture can vary by breed, body size, camera angle, and lighting; training and evaluation data needs to reflect that diversity, not just a convenient subset.
- **Human-in-the-loop by design:** AEIS is built to surface and explain a likely emergency, not to authorize or trigger any physical intervention on its own. A human is expected to confirm before any rescue action is taken.

## 11. Roadmap

The path from today's research vision to a deployed system runs through five stages. Stages 1 and 2 are in active development; Stages 3 through 5 are upcoming.

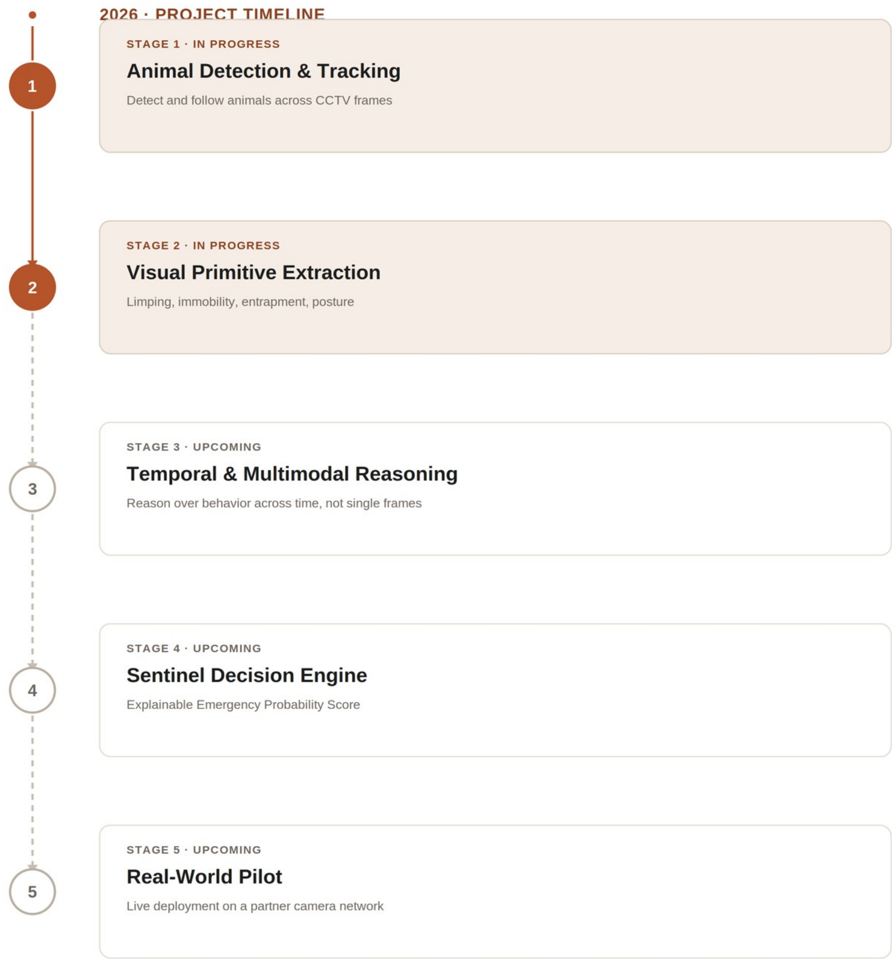


*Figure 2. Project roadmap from current research (Stages 1 and 2) to a real-world pilot (Stage 5).*

## 12. Long-Term Vision

The long-term goal is foundational AI infrastructure for understanding animal emergencies from existing video infrastructure, using computer vision, temporal reasoning, and multimodal AI not as a one-off detector, but as a general-purpose layer that any CCTV network could plug into.

## 13. Status & Next Steps

This document presents the research vision, the cascade architecture, and the evaluation plan. Implementation work is underway on the Stage 1 trigger model and the visual primitive extraction pipeline; multimodal reasoning integration and large-scale evaluation are the next milestones.

Implementation note: the initial prototype will use Gemini as the Stage 2 multimodal reasoning backbone. The architecture itself is model-agnostic. Stage 2 can be re-pointed at a different multimodal model as the field moves, without changing the cascade design, the visual primitives, or the SDE.

## 14. References

Foundational work this architecture builds on:

**Systems addressing related problems (Section 2):**

Norouzzadeh, M. S., et al. (2018). Automatically identifying, counting, and describing wild animals in camera-trap images with deep learning. PNAS. (Camera-trap species classification)

Mathis, A., et al. (2018). DeepLabCut: markerless pose estimation of user-defined body parts with deep learning. Nature Neuroscience. (Animal pose estimation)

Hasan, M., et al. (2016). Learning Temporal Regularity in Video Sequences. CVPR. (Generic video anomaly detection)

**Architecture components this design builds on:**

Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). You Only Look Once: Unified, Real-Time Object Detection. CVPR. (Object detection)

Wojke, N., Bewley, A., & Paulus, D. (2017). Simple Online and Realtime Tracking with a Deep Association Metric. ICIP. (Multi-object tracking)

Radford, A., et al. (2021). Learning Transferable Visual Models From Natural Language Supervision (CLIP). ICML. (Vision-language models)

Alayrac, J.-B., et al. (2022). Flamingo: a Visual Language Model for Few-Shot Learning. NeurIPS. (Vision-language models)

Carreira, J., & Zisserman, A. (2017). Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset. CVPR. (Temporal action recognition)

Google Gemini Team. (2023). Gemini: A Family of Highly Capable Multimodal Models. arXiv preprint. (Initial prototype backbone)